

BS-PH101 — Physics-I

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Module 3 Comprehensive Exam Notes: Electromagnetism & Materials

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1. Maxwell's Equations and Electromagnetic Waves

1.1 Maxwell's Four Fundamental Equations

Maxwell unified electricity and magnetism into four differential and integral equations governing electromagnetic phenomena.

Maxwell's Equations in Differential and Integral Forms

1. **Gauss's Law for Electrostatics:** Net electric flux through a closed surface equals charge enclosed:

$$\nabla \cdot \vec{D} = \rho \iff \oint_S \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}} \quad (\vec{D} = \epsilon_0 \vec{E} + \vec{P}) \quad (1)$$

2. **Gauss's Law for Magnetostatics:** Magnetic monopoles do not exist (magnetic flux lines are continuous loops):

$$\nabla \cdot \vec{B} = 0 \iff \oint_S \vec{B} \cdot d\vec{A} = 0 \quad (2)$$

3. **Faraday's Law of Electromagnetic Induction:** Time-varying magnetic field produces an electric field:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \iff \oint_C \vec{E} \cdot d\vec{r} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{A} \quad (3)$$

4. **Ampere-Maxwell Law:** Magnetic fields are generated by electric currents and time-varying electric fields:

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \iff \oint_C \vec{H} \cdot d\vec{r} = \int_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{A} \quad (4)$$

1.2 Displacement Current and Wave Equation in Free Space

Maxwell modified Ampere's law ($\nabla \times \vec{H} = \vec{J}$) by taking the divergence of both sides. Since $\nabla \cdot (\nabla \times \vec{H}) = 0$, it required $\nabla \cdot \vec{J} = 0$, which violates the charge continuity equation $\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$ for time-varying fields.

Substituting $\rho = \nabla \cdot \vec{D}$ into continuity equation:

$$\nabla \cdot \vec{J} + \frac{\partial}{\partial t} (\nabla \cdot \vec{D}) = 0 \implies \nabla \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) = 0 \quad (5)$$

Thus, Maxwell introduced the **Displacement Current Density** $\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$.

Electromagnetic Wave Equation and Speed of Light

In free space ($\rho = 0, \vec{J} = 0, \mu = \mu_0, \epsilon = \epsilon_0$), taking curl of Faraday's law yields:

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B}) \implies \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad (6)$$

Since $\nabla \cdot \vec{E} = 0$, this simplifies to the standard 3D wave equation:

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{and} \quad \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \quad (7)$$

Comparing with $\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$, the wave velocity in vacuum is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}} = 2.998 \times 10^8 \text{ m/s} \quad (8)$$

1.3 Poynting Vector and Energy Flow

Definition: Poynting Vector

The **Poynting Vector** ($\vec{S}$) represents the directional energy flux density (rate of energy transfer per unit area) of an electromagnetic field:

$$\vec{S} = \vec{E} \times \vec{H} \quad \left(\text{in } \frac{\text{W}}{\text{m}^2} \right) \quad (9)$$

The time-averaged Poynting vector intensity for a plane wave is $I = \langle S \rangle = \frac{1}{2} c \epsilon_0 E_0^2$.

Solved Problem 1: Displacement Current in a Capacitor

Problem: A parallel-plate capacitor with circular plates of radius $R = 0.1 \text{ m}$ is being charged by a current $I = 2.0 \text{ A}$. Calculate: (a) Displacement current I_d , (b) Rate of change of electric field dE/dt .

Solution:

1. Total displacement current between plates equals conduction current in wires:
 $I_d = 2.0 \text{ A}$.
2. Plate area $A = \pi R^2 = \pi(0.1)^2 = 0.0314 \text{ m}^2$.
3. $I_d = \epsilon_0 A \frac{dE}{dt} \implies \frac{dE}{dt} = \frac{I_d}{\epsilon_0 A}$:

$$\frac{dE}{dt} = \frac{2.0}{(8.854 \times 10^{-12}) \times 0.0314} = 7.19 \times 10^{11} \text{ V/(m} \cdot \text{s)} \quad (10)$$

2. Dielectric Properties of Materials

2.1 Dielectric Vectors: Polarisation, Susceptibility, Permittivity

Dielectrics are electrical insulators that can be polarized by an applied electric field.

Fundamental Dielectric Relations

- **Electric Dipole Moment ($\vec{p}$):** $\vec{p} = q\vec{d}$ (directed from $-q$ to $+q$).
- **Polarisation Vector ($\vec{P}$):** Dipole moment per unit volume: $\vec{P} = N\vec{p}$.
- **Electric Displacement ($\vec{D}$):**

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \epsilon_r \vec{E} \quad (11)$$

- **Electric Susceptibility (χ_e):** $\vec{P} = \epsilon_0 \chi_e \vec{E} \Rightarrow \epsilon_r = 1 + \chi_e$.

Microscopic Polarisation Mechanisms:

Mechanism	Origin / Description	Temperature Dependence
Electronic (α_e)	Displacement of electron cloud relative to nucleus in an atom.	Independent of temperature (T).
Ionic (α_i)	Relative displacement of positive and negative ions in a crystal lattice.	Independent of temperature (T).
Orientalional (α_o)	Realignment of permanent molecular dipoles along external field.	Strongly temperature dependent ($\alpha_o \propto 1/T$).
Space-Charge (α_s)	Accumulation of free charge carriers at interfaces/grain boundaries.	Dependent on temperature and frequency.

2.2 Clausius-Mossotti Equation

Consider a solid dielectric where local field acting on an atom is the **Lorentz Field**:

$$E_{\text{local}} = E + \frac{P}{3\epsilon_0} \quad (12)$$

The induced dipole moment per atom is $p = \alpha E_{\text{local}}$, so total polarisation for N atoms per unit volume is:

$$P = Np = N\alpha E_{\text{local}} = N\alpha \left(E + \frac{P}{3\epsilon_0} \right) \quad (13)$$

Clausius-Mossotti Relation Derivation

Solving for P :

$$P \left(1 - \frac{N\alpha}{3\epsilon_0} \right) = N\alpha E \Rightarrow P = \frac{N\alpha E}{1 - \frac{N\alpha}{3\epsilon_0}} \quad (14)$$

Since $P = (\epsilon_r - 1)\epsilon_0 E$, substitute for P :

$$(\epsilon_r - 1)\epsilon_0 E = \frac{N\alpha E}{1 - \frac{N\alpha}{3\epsilon_0}} \Rightarrow \frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0} \quad (15)$$

This connects the macroscopic dielectric constant ϵ_r with the microscopic atomic polarizability α .

3. Magnetic Properties of Materials

3.1 Magnetic Vectors and Classification

Magnetic Field Relations

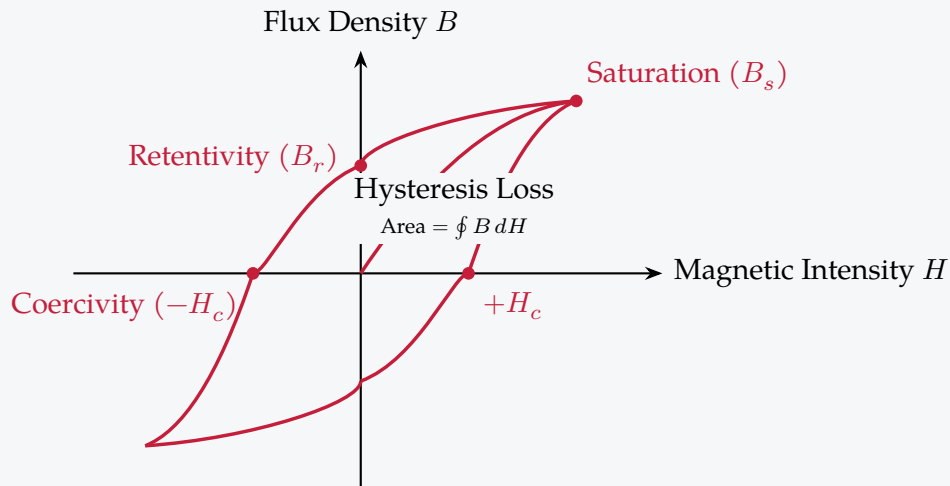
- **Magnetisation ($\vec{M}$):** Magnetic dipole moment per unit volume: $\vec{M} = \frac{\sum \vec{m}}{V}$.
- **Magnetic Flux Density ($\vec{B}$):** $\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_0\mu_r\vec{H}$.
- **Magnetic Susceptibility (χ_m):** $\vec{M} = \chi_m\vec{H} \Rightarrow \mu_r = 1 + \chi_m$.

Classification of Magnetic Materials:

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Atomic Dipoles	No permanent dipoles.	Permanent dipoles present (random).	Permanent dipoles strongly aligned in domains.
Susceptibility χ_m	Small and negative ($\sim -10^{-5}$).	Small and positive ($\sim 10^{-4}$).	Very large and positive ($\sim 10^3 - 10^5$).
Permittivity μ_r	$\mu_r < 1$ slightly.	$\mu_r > 1$ slightly.	$\mu_r \gg 1$.
Temp. Dependence	Independent of T .	Curie Law ($\chi \propto 1/T$).	Curie-Weiss Law ($\chi \propto \frac{1}{T-T_c}$).
External Field	Repelled by magnetic fields.	Feebly attracted to magnetic fields.	Strongly attracted into magnetic fields.
Examples	Bismuth, Copper, Water, Gold.	Aluminium, Platinum, Oxygen.	Iron, Cobalt, Nickel, Gadolinium.

3.2 Hysteresis Loop (B-H Curve)

Ferromagnetic Hysteresis (B-H) Loop



Hysteresis Terminology

- **Retentivity (Remanence B_r):** Residual flux density remaining in material when magnetising field H is reduced to zero.
- **Coercivity (H_c):** Reverse magnetising field required to reduce residual flux density B to zero.
- **Hysteresis Loss:** Energy dissipated per unit volume per cycle as heat during magnetisation reversal, equal to area enclosed by B - H loop: $W_{\text{loss}} = \oint B dH$.

Soft vs. Hard Magnetic Materials:

Feature	Soft Magnetic Materials	Hard Magnetic Materials
Hysteresis Area	Thin and narrow loop (low energy loss).	Broad and wide loop (high energy loss).
Coercivity	Very low coercivity (H_c small).	Very high coercivity (H_c large).
Retentivity	High retentivity, easily demagnetised.	High retentivity, retains magnetisation.
Applications	Transformer cores, electromagnets, inductors.	Permanent magnets, magnetic recording tapes, loudspeakers.

Solved Problem 2: Hysteresis Loss Power Dissipation

Problem: An iron ring of volume $V = 100 \text{ cm}^3$ is subjected to a 50 Hz alternating magnetic field. If the area of the hysteresis loop corresponds to 250 J/m^3 per cycle, calculate the power loss in watts.

Solution:

1. Volume $V = 100 \times 10^{-6} \text{ m}^3 = 10^{-4} \text{ m}^3$.
2. Frequency $f = 50 \text{ Hz}$ (50 cycles/second).
3. Energy loss per second (Power P) = Energy/cycle/volume $\times$ Volume $\times$ Frequency:

$$P = W_{\text{loss}} \times V \times f = 250 \text{ J/m}^3 \times 10^{-4} \text{ m}^3 \times 50 \text{ s}^{-1} = 1.25 \text{ W} \quad (16)$$

Quick Revision — Module 3

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Key Formulas and Metrics at a Glance

Maxwell 1 (Gauss Electro)	$\nabla \cdot \vec{D} = \rho \iff \oint_S \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}}$
Maxwell 2 (Gauss Magneto)	$\nabla \cdot \vec{B} = 0 \iff \oint_S \vec{B} \cdot d\vec{A} = 0$
Maxwell 3 (Faraday)	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \iff \oint_C \vec{E} \cdot d\vec{r} = -\frac{\partial \Phi_B}{\partial t}$
Maxwell 4 (Ampere-Maxwell)	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$
Displacement Current	$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$
Speed of Light in Vacuum	$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$
Poynting Vector	$\vec{S} = \vec{E} \times \vec{H} \quad (\text{W/m}^2)$
Clausius-Mossotti Eq.	$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$
Magnetic Relations	$\vec{B} = \mu_0(\vec{H} + \vec{M}) \quad \text{and} \quad \mu_r = 1 + \chi_m$
Hysteresis Power Loss	$P_{\text{loss}} = (\oint B dH) \times \text{Volume} \times \text{Frequency}$

One-Line Definitions

- **Displacement Current:** A quantity appearing in Maxwell's equations defined by rate of change of electric displacement field ($\vec{J}_d = \partial \vec{D} / \partial t$).
- **Poynting Vector:** A vector representing the directional energy flux density of an electromagnetic field ($\vec{S} = \vec{E} \times \vec{H}$).
- **Lorentz Local Field:** The actual local electric field acting at the position of an individual atom in a solid ($E_{\text{local}} = E + P/3\epsilon_0$).
- **Retentivity (B_r):** Residual magnetic flux density remaining in a ferromagnetic material when magnetising field H is reduced to zero.
- **Coercivity (H_c):** The magnitude of reverse magnetic field intensity required to reduce residual magnetisation to zero.

Frequently Asked / Viva Questions

Q1. Why did Maxwell modify Ampere's law and what term did he add?

Ampere's law ($\nabla \times \vec{H} = \vec{J}$) violated charge conservation for time-varying fields ($\nabla \cdot \vec{J} \neq 0$). Maxwell added displacement current density $\vec{J}_d = \partial \vec{D} / \partial t$.

Q2. What is the physical significance of $\nabla \cdot \vec{B} = 0$?

Frequently Asked / Viva Questions (contd.)

It signifies that isolated magnetic monopoles do not exist in nature and magnetic flux lines always form continuous closed loops.

Q3. What is Poynting vector and state its SI unit?

$\vec{S} = \vec{E} \times \vec{H}$ represents the rate of electromagnetic energy flow per unit area. Its SI unit is W/m^2 .

Q4. State the relation between dielectric constant ϵ_r and electric susceptibility χ_e .

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0(1 + \chi_e)\vec{E} = \epsilon_0 \epsilon_r \vec{E} \implies \epsilon_r = 1 + \chi_e.$$

Q5. Derive Clausius-Mossotti relation connecting macroscopic and microscopic dielectric properties.

Using Lorentz local field $E_{\text{local}} = E + P/3\epsilon_0$, total polarisation $P = N\alpha E_{\text{local}} = N\alpha(E + P/3\epsilon_0)$. Substituting $P = (\epsilon_r - 1)\epsilon_0 E$ yields $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$.

Q6. Distinguish between polar and non-polar dielectrics.

Polar dielectrics have permanent molecular dipole moments (e.g. H_2O , HCl); non-polar dielectrics have zero permanent dipole moment in zero field (e.g. O_2 , N_2 , CO_2).

Q7. Why does orientational polarizability decrease with increasing temperature?

Because higher thermal agitation randomizes the orientation of permanent molecular dipoles, opposing their alignment along the external field ($\alpha_o \propto 1/T$).

Q8. Distinguish between diamagnetic and paramagnetic materials.

Diamagnetic materials have no permanent dipoles, negative susceptibility ($\chi_m < 0$), and are repelled by magnetic fields. Paramagnetic materials have permanent dipoles, positive susceptibility ($\chi_m > 0$), and are feebly attracted.

Q9. State Curie-Weiss law for ferromagnetic materials.

Above the Curie temperature T_c , a ferromagnetic material becomes paramagnetic with susceptibility $\chi_m = \frac{C}{T - T_c}$.

Q10. What is meant by hysteresis loop area?

The area enclosed by the B - H loop represents the energy dissipated as heat per unit volume per magnetization reversal cycle ($W_{\text{loss}} = \oint B dH$).

Q11. Which magnetic material is suitable for transformer cores and why?

Soft magnetic materials (e.g. Soft Iron, Silicon Steel) because they have low coercivity and thin hysteresis loops, minimizing energy loss per cycle.

Q12. Show that speed of EM wave in vacuum is $c = 1/\sqrt{\mu_0 \epsilon_0}$.

From Maxwell's free-space wave equation $\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$, comparing with standard wave equation $\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$ yields $v = c = 1/\sqrt{\mu_0 \epsilon_0} \approx 3 \times 10^8 \text{ m/s}$.

Summary Comparison of Dielectric Polarization Mechanisms

Mechanism	Displacement Type	Frequency Range	Temperature Dependence
Electronic (α_e)	Electron cloud relative to nucleus.	Optical ($\sim 10^{15}$ Hz).	Independent of T .
Ionic (α_i)	Positive and negative ion displacement.	Infrared ($\sim 10^{13}$ Hz).	Independent of T .
Orientational (α_o)	Alignment of permanent dipoles.	Microwave ($\sim 10^9$ Hz).	Inverse with T ($\alpha_o \propto 1/T$).
Space-Charge (α_s)	Migration of charge carriers.	Audio ($\sim 10^3$ Hz).	Complex T dependence.